PROJECT TITLE

SAHAYAK – An AI-Powered External Memory Companion for Alzheimer’s & Dementia Patients

An autonomous, wearable AI agent that continuously Observes the environment.

TEAM DETAILS

Team Name: Percevia

Team Members:
- Yash Goel– Team Leader
- Tanish Aggarwal
- **Khushi Sharma**
- **Palak Gupta**

College: Vivekananda Institute Of Professional Studies, VIPS-TC

PROBLEM DOMAIN **:- Open innovation (Healthcare & IOT)**

THE CORE PROBLEM: **THE "CONTEXT GAP" & LOSS OF INDEPENDENCE**

Alzheimer’s and **Dementia** are not merely about forgetting names; they strip away a person's ability to recall the narrative of life, leading to a complete loss of independence.

Patients suffer from **Episodic Memory Loss**, meaning they forget the context of recent events, which manifests in three specific ways:

1. **Loss of Object Permanence (Short-Term Lapses):**
   - Patients constantly struggle with questions like "Where did I put my glasses?" or "Where are my keys?".
   - This leads to anxiety and helplessness in daily tasks.
2. **Social Disconnection (Prosopagnosia / Face Blindness):**
   - They may recognize a face but fail to recall the context: "I know him, but is he my son or my neighbor?".
   - This causes social withdrawal and isolation.
3. **Action & Safety Uncertainty (Visual Agnosia):**
   - Action Verification: Forgetting critical health tasks like "Did I already take my medicine?".
   - Safety Risks: Inability to interpret objects correctly, such as confusing a bottle of cleaning fluid for a drink.

Limitations of Existing Solutions:

• **Smart Speakers (Alex a/Siri):** They are connected to the internet but la
They can answer general facts but cannot answer personal questions like "Where is *my* wallet?".

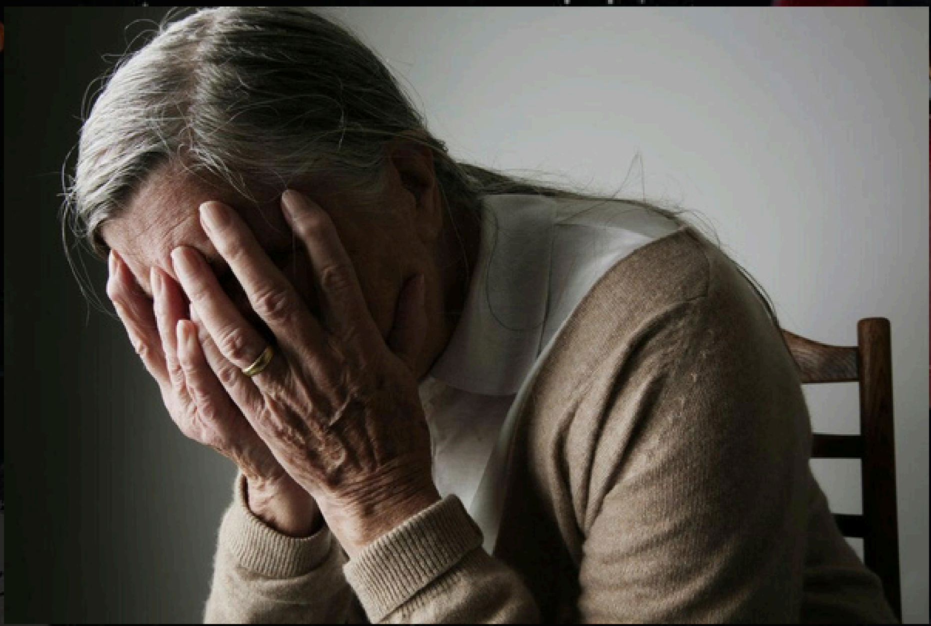
• **GPS Trackers:** They only indicate **location** (where the user is), failing to explain **activity** (what the user is doing).

• **Reminder Apps:** These are **passive** and fundamentally flawed because they require the patient to remember to input the data first.

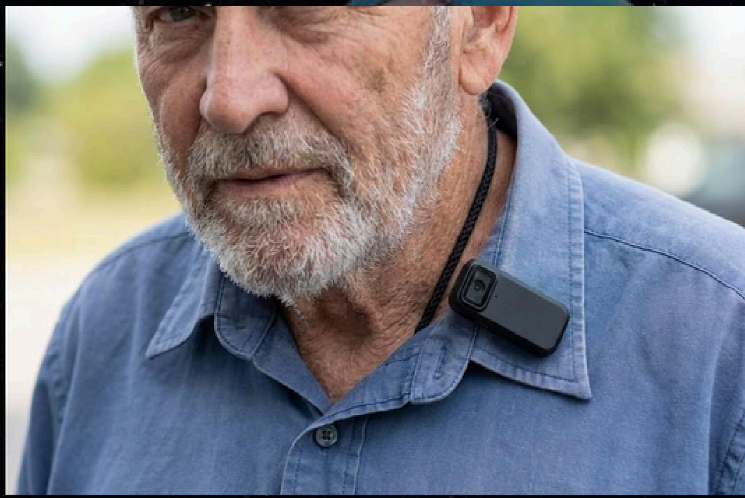
• **CCTV Cameras:** They provide **passive recording** without intelligence, meaning they cannot process video in real-time or answer user queries.

• **Complex ity:** Many existing apps are too difficult for elderly users to nav
need active, voice-guided assistance.

• **Symptom Focus:** Current tools only provide **alerts** rather than actually **recreating the lost memory**.



Proposed Solution: SAHAYAK (THE EXTERNAL HIPPOCAMPU



3.1. High-Level Overview

Sahayak is an autonomous, wearable AI agent designed to function as an **"Ex Hippocampus"** for patients suffering from Alzheimer's and Dementia. Unlike passive GPS trackers or generic smart speakers, Sahayak actively observes the user's environment to create an **Artificial Episodic Memory**. It continuously logs "events" (who, what, when, where) and retrieves this context via voice commands to restore the user's independence.

3.2. Core Philosophy: The Split-Module Architecture

To balance high-performance AI processing with wearer comfort, the solution utilizes a unique **"Split-Design"** architecture:

• **Module A: The "Eye" (Neural Clip):** A lightweight (<40g), fabric-covered camera and microphone unit worn on the collar. It is designed to look like an accessory, not a medical device.

• **Module B: The "Brain" (Core Unit):** A processing unit (Raspberry Pi 4/5 + Battery) kept in the pocket or a belt pouch. This ensures heat and weight remain away from the user's body.

• **The Link:** A concealed ribbon cable connects the sensory "Eye" to the processing "Brain".

3.3. Technical Implementation Phases

The development is structured in two distinct phases to ensure stability and scalability:

•**Phase 1: Prototype (Laptop-Based):** utilizes a laptop's CPU/GPU for heavy inference (YOLOv8 + CLIP) with a webcam and microphone to validate the software logic.

•**Phase 2: Wearable (Edge-Based):** Migrates the stack to a **Raspberry Pi 4 Model B (4GB)** with a Pi Camera Module 3 and **Bone Conduction Headphones** for private audio output.

SYSTEM ARCHITECTURE (HARDWARE CENTRIC)

4.1. The "Split-Module" Design Philosophy

To address the challenge of running heavy AI models (like YOLO and CLIP) on a wearable device without causing overheating or discomfort, Sahayak employs a **Split-Module Architecture**. This design physically separates the lightweight sensory components (The "Eye") from the heavy processing and power components (The "Brain").

4..2. Module A: The "Neural Clip" (Sensory Unit)

•**Placement:** Worn as a magnetic clip on the shirt collar or chest area.
•**Function:** Acts as the device's "eyes and ears," capturing real-time environmental data.
•**Key Components:**

- •**Visual Sensor: Raspberry Pi Camera Module 3** (Wide Angle) is used to capture high-resolution frames for the vision pipeline.
- •**Audio Input:** A **MEMS Microphone** (or USB Mic) captures natural language queries from the user.
- •**Interaction:** Features a tactile **Wake Button** allowing the user to manually trigger the system.

•**Form Factor:** Designed with a "Pebble" aesthetic and fabric cover, weighing less than 40g, to resemble a fashion accessory rather than a medical device.

4.3. Module B: The "Core Unit" (Processing Unit)

•**Placement:** Concealed in the user's pocket or a belt pouch.
•**Function:** Handles all heavy Edge AI processing, power management, and heat dissipation away from the sensitive chest area.
•**Key Components:**

- •**Compute Engine: Raspberry Pi 4 Model B (4GB RAM)** (or Pi 5) serves as the central brain handling offline AI inference.
- •**Power Source:** A **10,000mAh Battery** ensures 12+ hours of continuous operation.
- •**Telemetry:** Includes a **GPS Module** for location tracking and a specialized **Heat Dissipation Case** to manage thermal output.

4.4. Interconnectivity & Output

•**The Link:** A thin, concealed ribbon cable runs under the user's shirt, connecting the "Neural Clip" (Eye) to the "Core Unit" (Brain).

•**Audio Output:** The system utilizes **Bone Conduction Headphones** connected via Bluetooth.

•*Advantage:* These sit on the cheekbones, keeping the ear canal open to ambient sounds, ensuring the user stays aware of their surroundings while receiving private audio cues

FEATURES:-

1.Contextual Social Anchor (Face + History)

•**Function:** The system identifies people and recalls the *context* of the relationship rather than just the name.

•**Output:** Instead of saying "Person Detected," it whispers, *"This is your son, Rahul. He last visited you on Tuesday"*.

•**Benefit:** Addresses **Prosopagnosia** (face blindness) and reduces social anxiety during interactions.

2. Retroactive Memory Search (The Object Finder)

•**Function:** The device passively logs the location of essential items (Keys, Wallet, Glasses) into an offline database using visual object detection.

•**Output:** When the user asks *"Where are my glasses?"*, the system retrieves the last seen location: *"I saw them 15 minutes ago on the kitchen table"*.

•**Benefit:** Solves the issue of **Short-Term Memory Lapses** and object permanence.

3. The "Reality Scanner" (Visual Question Answering)

•**Function:** Uses OCR (Optical Character Recognition) and Visual Question Answering to interpret the safety of objects.

•**Output:** Can read expiry dates or identify dangerous items, warning the user: *"This is a bottle of bleach. Do not drink it"*.

•**Benefit:** Mitigates **Visual Agnosia** and safety risks in daily life.

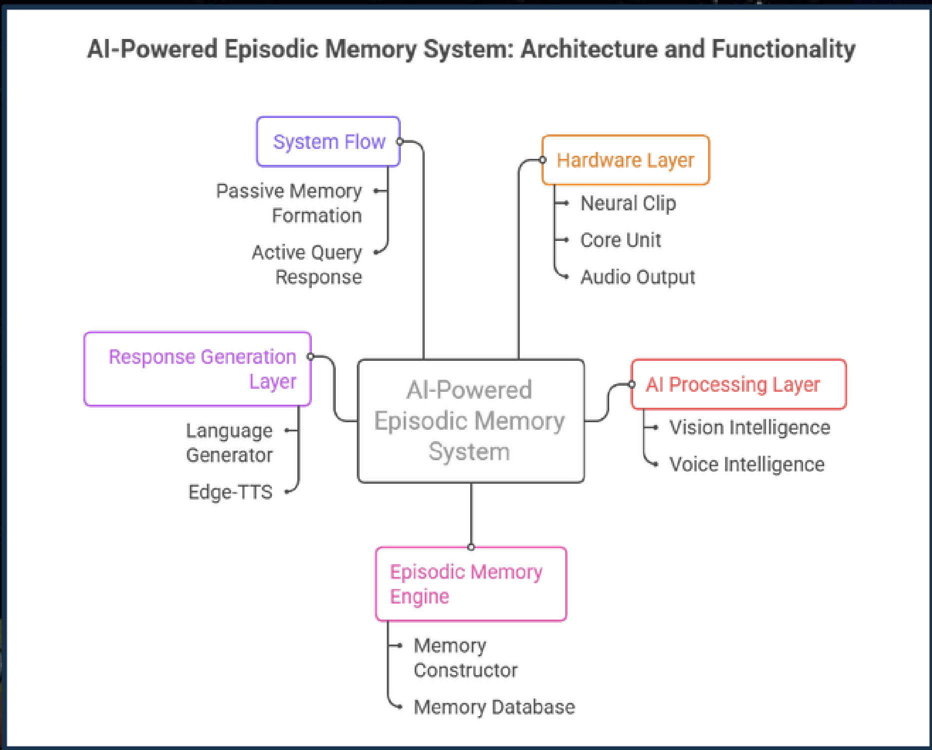
4. Action Verification & Logging

Function: The system tracks specific actions to verify if a task was completed .
Output: Answers queries like *"Did I take my medicine?"* with specific confirmations: *"Yes, you took the blue pill at 2 PM"* .
Benefit: Prevents accidental overdose or missed medication due to memory gaps.

5. Anxiety Reduction (Voice Cloning)

Function: Utilizes a Text-to-Speech (TTS) engine trained on a loved one's voice (e.g., a family member) instead of a robotic AI voice.
Benefit: Provides emotional reassurance and reduces the patient's fear of using technology.
6. Privacy-First "Shadow" Architecture
Function: All data processing (Vision, Voice, Logic) happens 100% offline on the Raspberry Pi .
Benefit: Ensures **Data Sovereignty**—personal video feeds and memories are never uploaded to the cloud .

Architecture and workflow diagram



1.Input & Perception

Visual: The Camera continuously scans the environment (video feed).
Audio: The Microphone listens for the "Wake Word" or user commands.

2. AI Processing (The "Brain")

Object Detection: YOLOv8 identifies objects (e.g., "Keys", "Medicine").
Identity Matching: FaceNet recognizes people (e.g., "Son", "Doctor").
Contextual Understanding: CLIP verifies if the object is the *same* specific item seen before (e.g., "My specific red cup").

3. Episodic Memory Creation

Stability Check: The system confirms the object is placed (stationary) and not just moving.
Logging: It saves a structured memory log locally: {Object: "Glasses", Location: "Kitchen Table", Time: "10:00 AM"}.

4. Query Processing

User Question: User asks, *"Where are my glasses?"*
Speech-to-Text: OpenAI Whisper converts the voice into text.
Search: The system scans the local database for the latest entry of "Glasses."

5. Response & Action

Answer Generation: The system formulates a natural sentence: *"You left them on the Kitchen Table."*
Audio Output: Edge-TTS speaks the answer via **Bone Conduction Headphones**.

Advantages of Sahayak

- Fully on-device AI
- Works offline
- Preserves user privacy
- Human-like memory recall
- Wearable & scalable
- Assistive, not intrusive

Applications

- Alzheimer's patients
- Dementia care
- Elderly assistance
- Rehabilitation centers
- Assisted living homes

Limitations

- Limited by Raspberry Pi computer
- Accuracy depends on environment
- Requires calibration per user

Future Scope

- Persistent memory database
- Bone-conduction audio
- Multi-language support
- Caregiver mobile app
- Cloud sync (optional)
- Emotion-aware responses